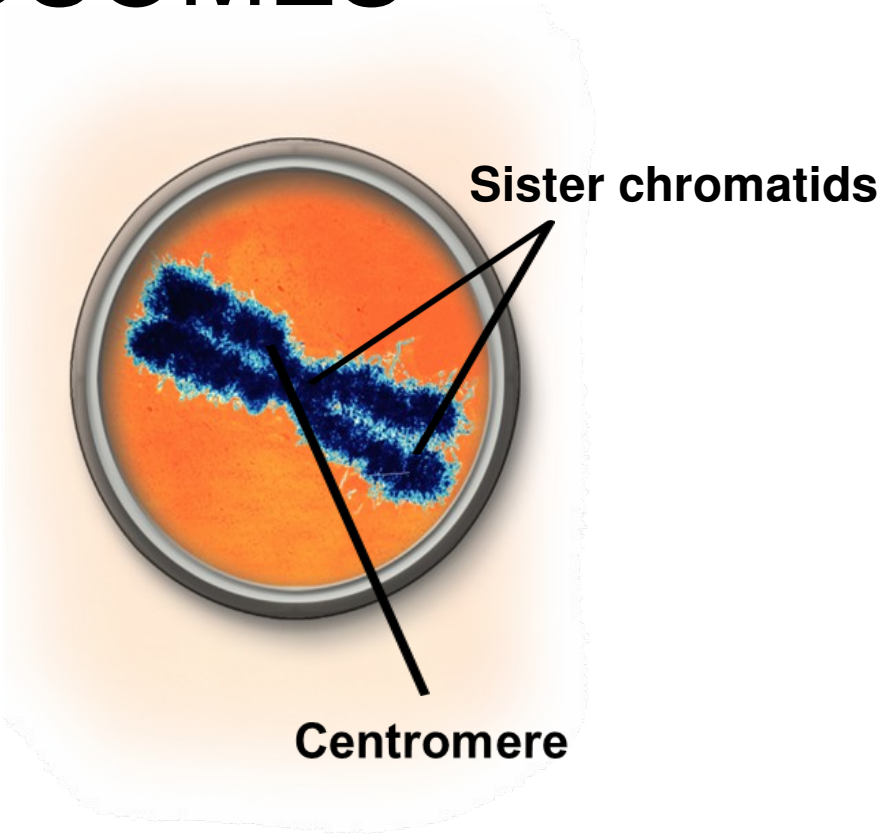
THE CELL CYCLE AND MITOSIS (2)

CHROMOSOMES

Each chromosome consists of two identical “sister” chromatids.

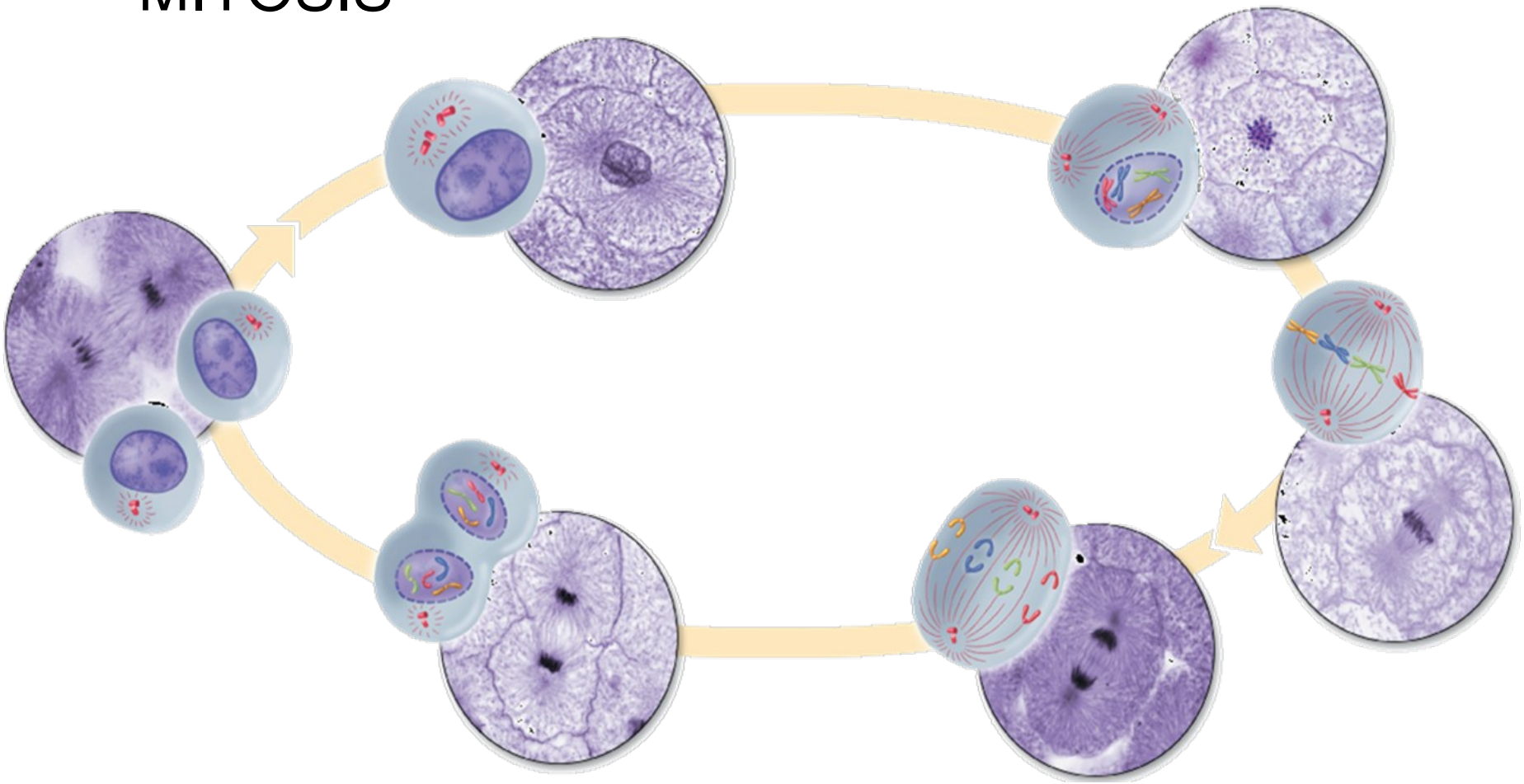
Each pair of chromatids is attached at an area called the centromere.



MITOSIS – CELL DIVISION

- Occurs in body cells – somatic cells
- Forms two identical daughter cells – exactly like the original
- Biologists divide the events of mitosis into four phases: (PMAT)
  - Prophase
  - Metaphase
  - Anaphase
  - Telophase

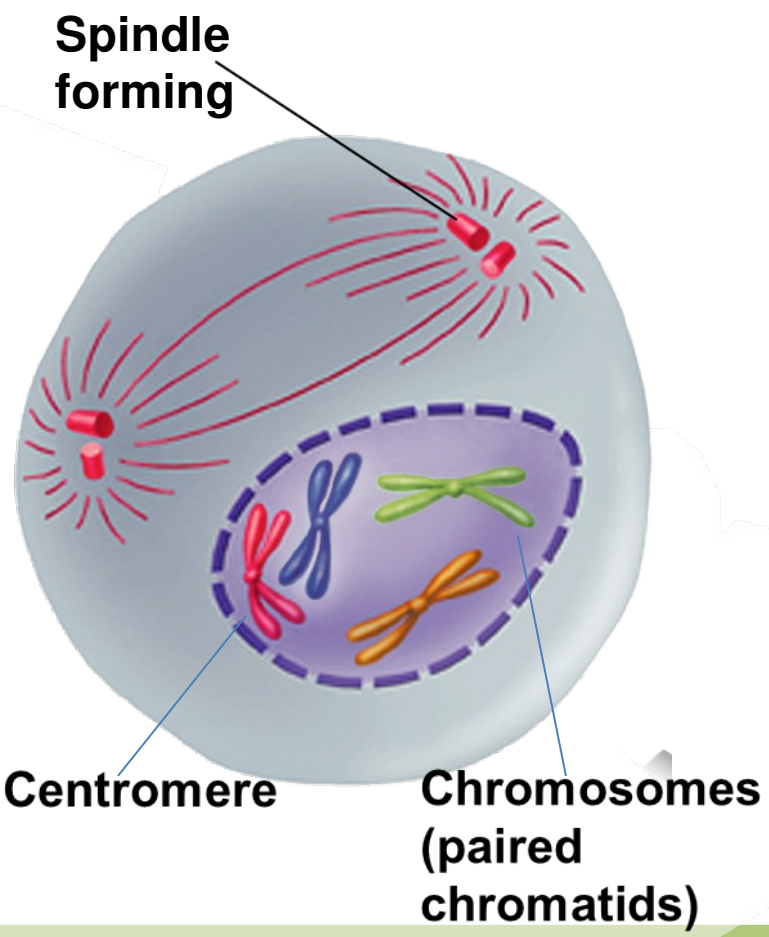
MITOSIS



PROPHASE

Prophase is the first and longest phase of mitosis.

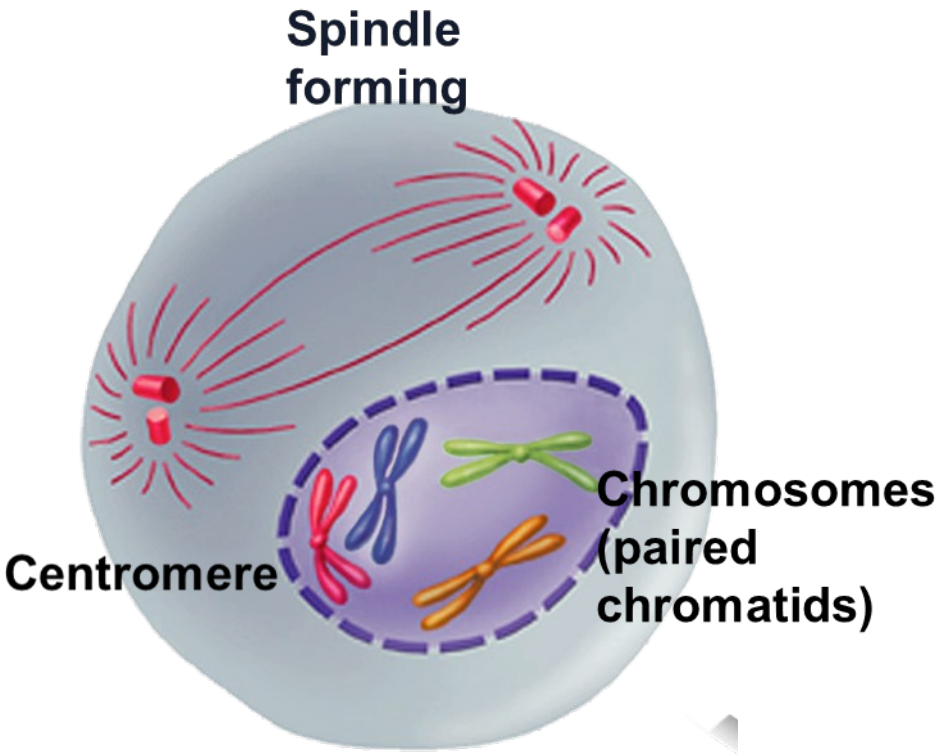
The centrioles separate and take up positions on opposite sides of the nucleus.



PROPHASE

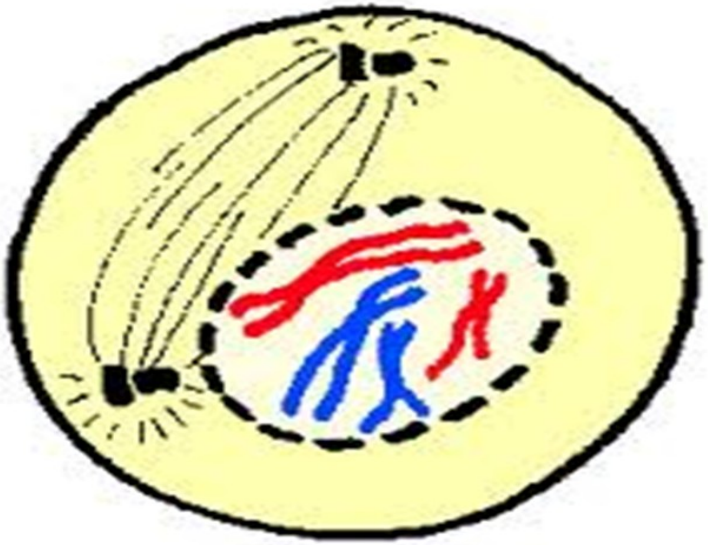
The centrioles lie in a region called the centrosome.

The centrosome helps to organize the spindle, a fanlike microtubule structure that helps separate the chromosomes.



SUMMARY OF PROPHASE

- First phase of MITOSIS (Longest Phase)
- Chromosomes become visible.
- Centrioles separate and move to opposite poles (ends).
- Spindle fibers appear.
- Nucleolus and nuclear membrane (envelope) disappear.

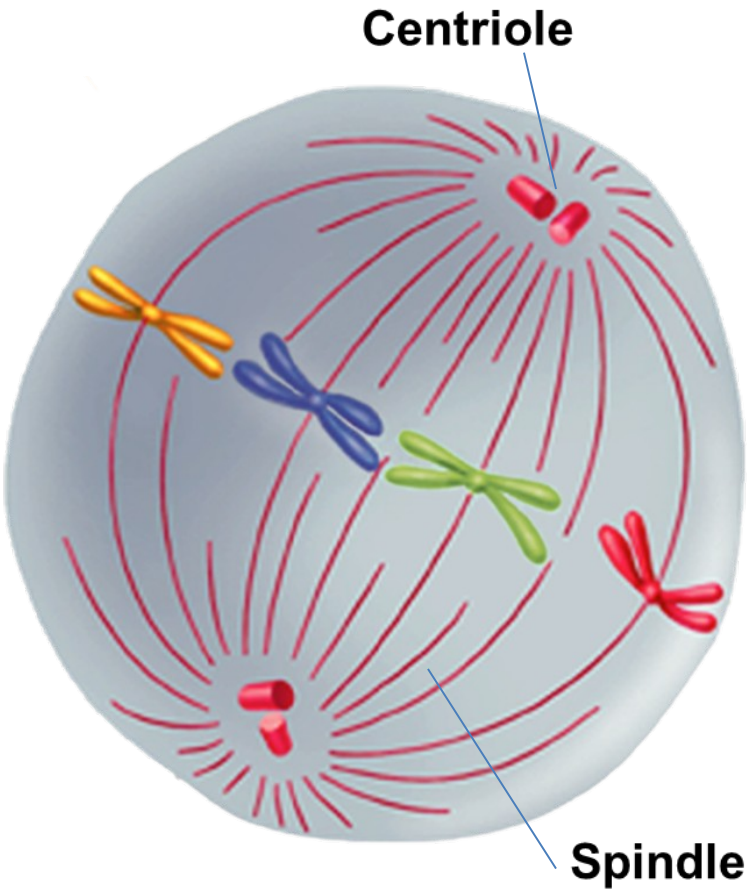


METAPHASE

The second phase of mitosis is metaphase.

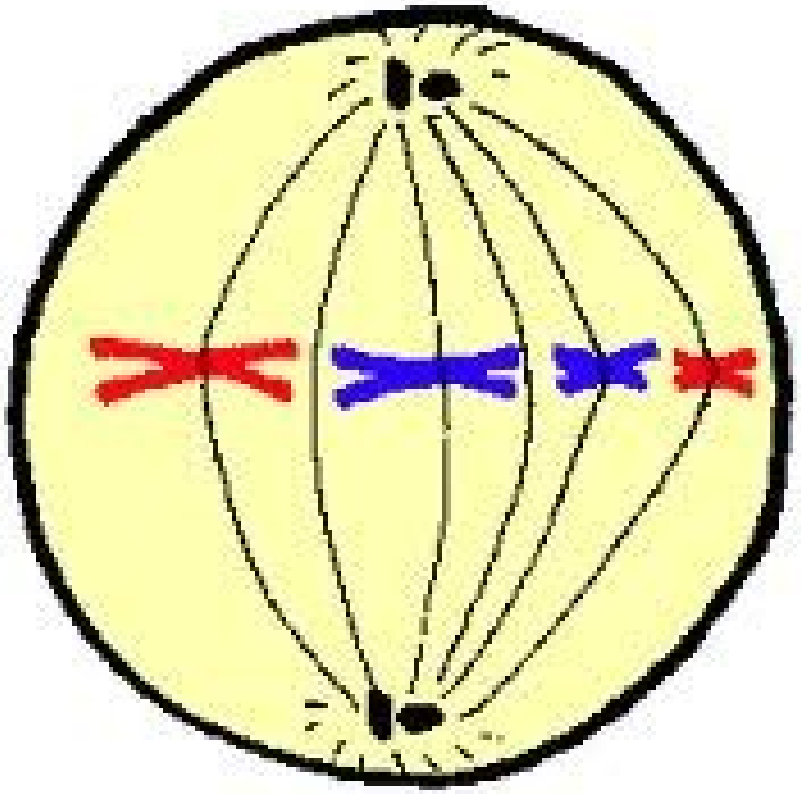
The chromosomes line up across the center of the cell.

Microtubules connect the centromere of each chromosome to the poles of the spindle



SUMMARY OF METAPHASE

- Shortest Phase
- Chromosomes attach to spindle fibers in center of the cell

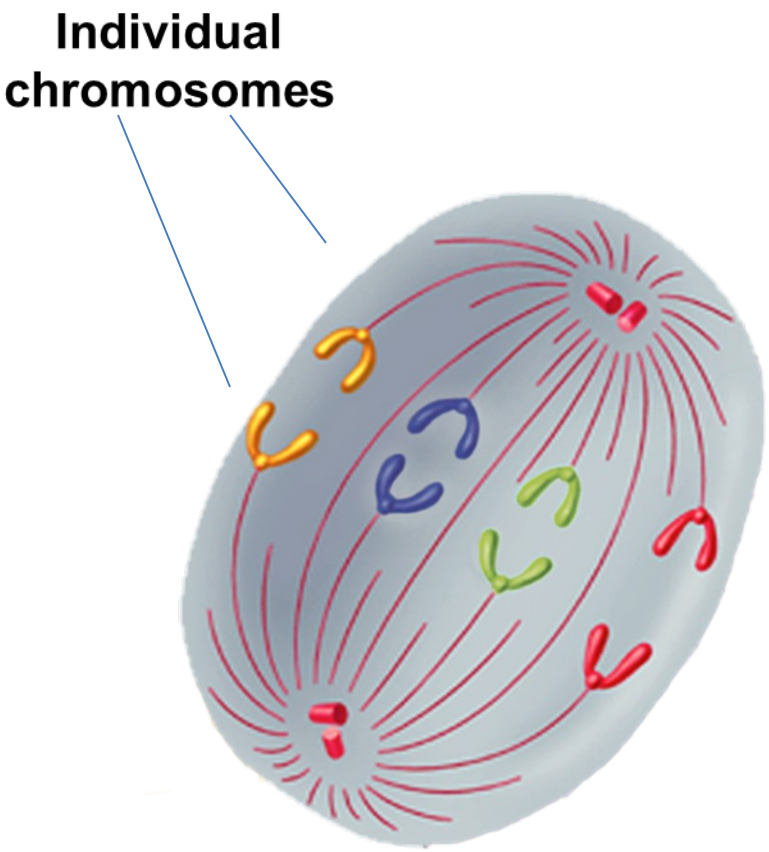


Anaphase

The third phase of mitosis.

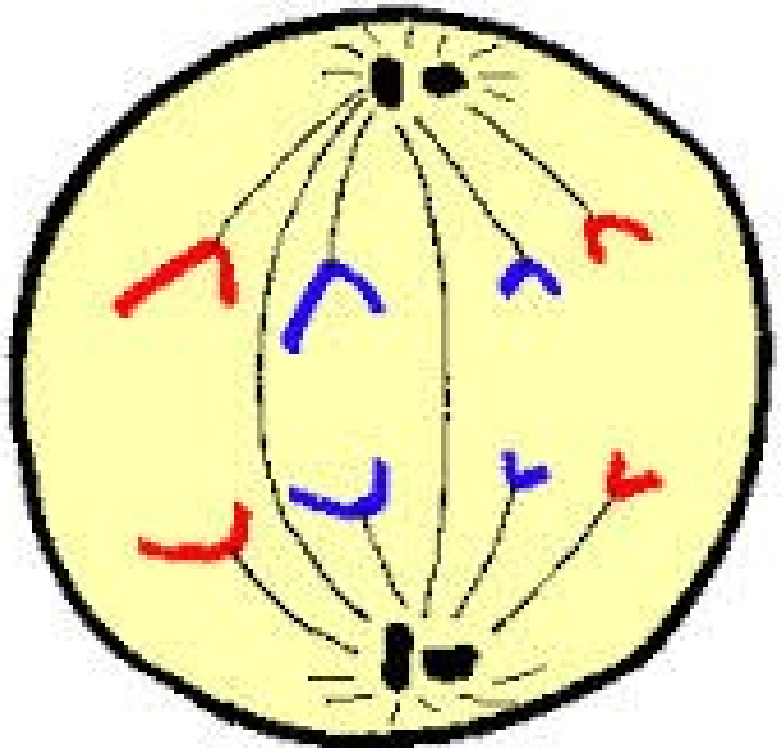
The sister chromatids separate into individual chromosomes.

The chromosomes continue to move until they have separated into two groups.



SUMMARY OF ANAPHASE

- Centromeres split apart and chromatids separate from one another.
- Each chromatid moves to opposite poles.

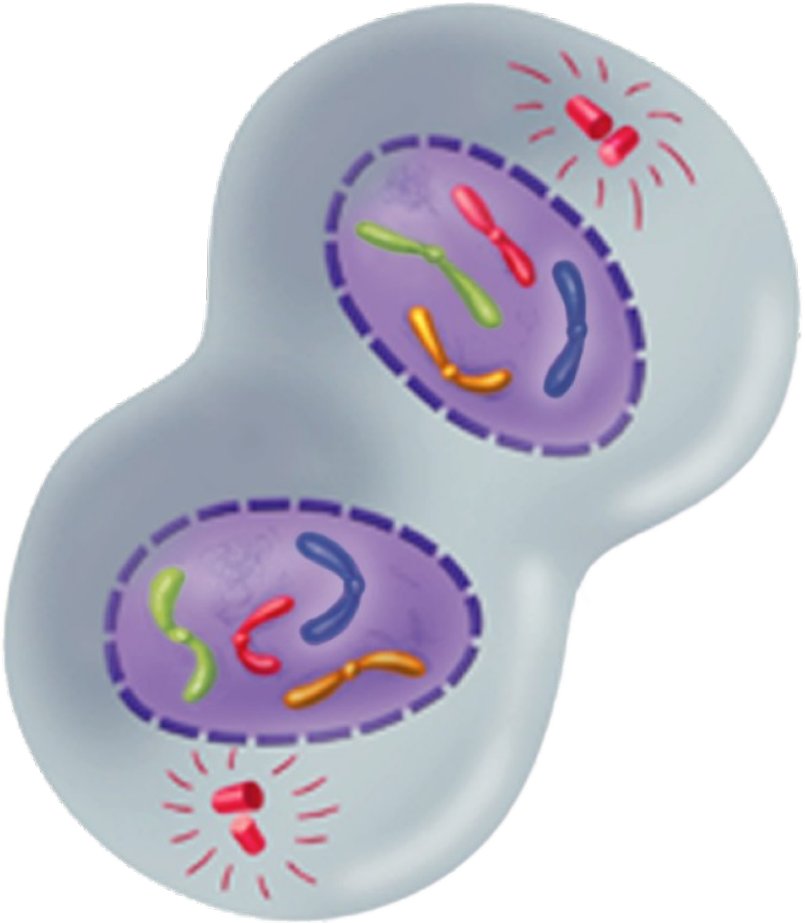


TELOPHASE

The fourth and final phase of mitosis.

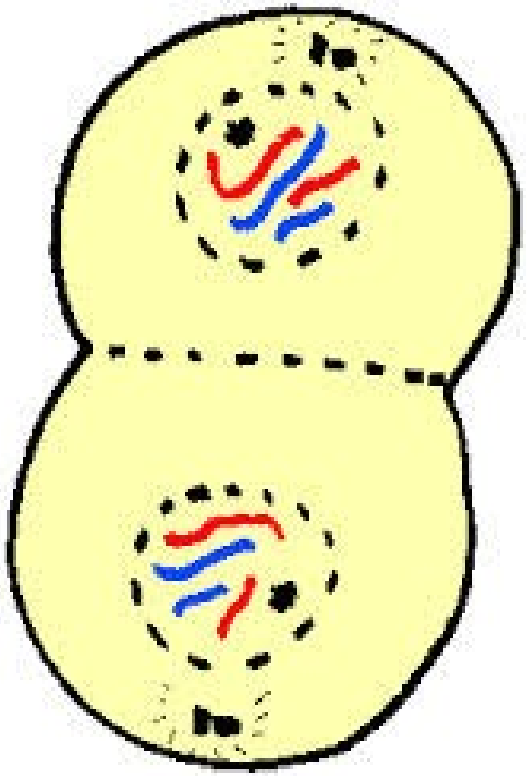
Chromosomes gather at opposite ends of the cell and lose their distinct shape

A new nuclear envelope forms around each cluster of chromosomes



SUMMARY OF TELOPHASE

- Chromosomes reach opposite poles of cell
- Chromatids unwind back into chromatin
- Nuclear envelope and nucleolus reappear reforming the nucleus
- Spindle fibers disappear
- New double membrane n between 2 nuclei (cell pinches)
- Animal cell – cleavage
- Plant cell – cell plate



CYTOKINESIS – CYTOPLASM DIVISION

During cytokinesis, the cytoplasm pinches in half.

Each daughter cell has an identical set of duplicate chromosomes